The polluted atmosphere as a shallow domain

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The shallow domain concept

- Natural, large-scale geophysical fluids share the shallow domain characteristics
- "almost" two-dimensional set → we can significantly simplify the model in the vertical direction
- a domain that is excessively greater horizontally than in the vertical dimension → leading to the classic hydrostatic approximation of the Navier-Stokes equations

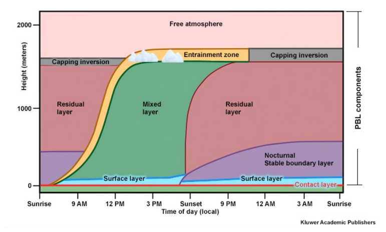

Fig. 1: A local slice of the atmosphere. Picture credit: mountwashington.org

The main ideas and tools in this scenario:

- introduce the aspect ratio $\epsilon = \frac{\text{characteristic depth}}{\text{characteristic width}}$,
- anisotropic viscosity hypothesis,
- rescaling, $x = x_1, y = x_2, z = \epsilon x_3$.

Correspondingly, we rescale

- the kinematic terms: $v_3 = \epsilon u_3^\epsilon$,
- the viscosity terms: $\nu_z = \epsilon^2 \nu_3$,
- the diffusivity terms: $K_x = \epsilon K_1, K_y = K_2, K_z = \epsilon^2 K_3$.

The anisotropic equations

$$\partial_t u_1^\epsilon + \mathbf{u}^\epsilon \cdot \nabla u_1^\epsilon - \Delta_\nu u_1^\epsilon - \alpha u_2^\epsilon + \epsilon \beta u_3^\epsilon + \partial_1 p^\epsilon = 0 \text{ in } \Omega \times (0,T)$$

$$\partial_t u_2^\epsilon + \mathbf{u}^\epsilon \cdot \nabla u_2^\epsilon - \Delta_\nu u_2^\epsilon + \alpha u_1^\epsilon + \partial_2 p^\epsilon = 0 \text{ in } \Omega \times (0,T)$$

$$\epsilon^2 \{\partial_t u_3^\epsilon + \mathbf{u}^\epsilon \cdot \nabla u_3^\epsilon - \Delta_\nu u_3^\epsilon\} - \epsilon \beta u_1^\epsilon + \partial_3 p^\epsilon = 0 \text{ in } \Omega \times (0,T)$$

$$\nabla \cdot \mathbf{u}^\epsilon = 0 \text{ in } \Omega \times (0,T)$$

$$\partial_t C^\epsilon + \mathbf{u}^\epsilon \cdot \nabla C^\epsilon = \epsilon K_1 \frac{\partial^2 C^\epsilon}{\partial x_1^2} + K_2 \frac{\partial^2 C^\epsilon}{\partial x_2^2} + K_3 \frac{\partial^2 C^\epsilon}{\partial x_3^2} + S_\epsilon \text{ in } \Omega \times (0,T)$$

$$\mathbf{u}^\epsilon = 0 \text{ on } \Gamma \times (0,T)$$

$$\mathbf{u}^\epsilon(\cdot, t=0) = \mathbf{u}_0, C^\epsilon(\cdot, t=0) = C_0^\epsilon \text{ in } \Omega$$

$$C^\epsilon = 0 \text{ on } (\Gamma_U \cup \Gamma_L) \times (0,T), \partial_2 C^\epsilon = \partial_3 C^\epsilon = 0 \text{ on } \Gamma_G \times (0,T), \partial_1 C^\epsilon \in L^2 L^2(0,T;\Gamma_G)$$

The original equations We use the incompressible Navier-Stokes equations in all three space dimensions for the fluid velocity.

The limit model

In the vertical dimension we arrive to the hydrostatic approximation.

$$\partial_t u_1 + \mathbf{u} \cdot \nabla u_1 - \Delta_\nu u_1 - \alpha u_2 + \partial_1 p = 0 \text{ in } \Omega \times (0,T)$$

$$\partial_t u_2 + \mathbf{u} \cdot \nabla u_2 - \Delta_\nu u_2 - \alpha u_1 + \partial_2 p = 0 \text{ in } \Omega \times (0,T)$$

$$\partial_3 p = 0 \text{ in } \Omega \times (0,T)$$

$$\nabla \cdot \mathbf{u} = 0 \text{ in } \Omega \times (0,T)$$

$$\partial_t C + \mathbf{u} \cdot \nabla C = K_2 \frac{\partial^2 C}{\partial x_2^2} + K_3 \frac{\partial^2 C}{\partial x_3^2} + S \text{ in } \Omega \times (0,T)$$

$$u_1 = u_2 = u_3 n_3 = 0 \text{ on } \Gamma \times (0,T)$$

$$u_i(\cdot, t=0) = u_{0i}, C(\cdot, t=0) = C_0 \text{ in } \Omega, i = 1,2$$

$$C = 0 \text{ on } (\Gamma_U \cup \Gamma_L) \times (0,T), \partial_2 C = \partial_3 C = 0 \text{ on } \Gamma_G \times (0,T), \partial_1 C \in L^2 L^2(0,T;\Gamma_G)$$

The main theorem

- We provide an expansion of the convergence result of [Azérad and Guillén, 2001].
- The main theorem of their article is for the case of the Navier-Stokes equations of the ocean, without the inclusion of salinity.
- We prove that the hydrostatic approximation is still justifiable with the additional pollution concentration effect.
- The theorem is valid locally (see graph below).

Theorem

Let $\mathbf{u}_0 \in L^2(\Omega)^3$, with $\nabla \cdot \mathbf{u}_0 = 0, \mathbf{u}_0 \cdot \mathbf{n} = 0$ on $\partial\Omega$; and let $C(0) \in L^2(\Omega)^3$, $C(0) = 0$ on $\partial\Omega$; there exists a weak solution $(\mathbf{u}, \mathrm{C})$ of the hydrostatic equations of the polluted atmosphere, obtained as a limit of weak solutions $(\mathbf{u}^{\text{fl}}, C^\epsilon)$ of the anisotropic equations as the aspect ratio ϵ tends to 0.

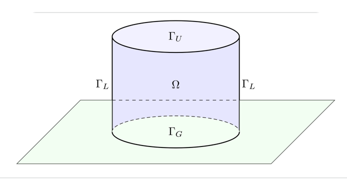

Fig. 2: A local slice of the atmosphere

A priori estimates, weak convergence

Fanciness:

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Compactness

Fanciness: Let's see some figures.

There are also callout blocks that allow for a more interesting layout of the poster.

\calloutblock[rotate angle]{from coordinate}{coordinate}{Block Width}{Block Content}

The alias for such blocks is *note*.

Existence

Fanciness:

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References

[Azérad and Guillén, 2001] Azérad, P. and Guillén, F. (2001). Mathematical justification of the hydrostatic approximation in the primitive equations of geophysical fluid dynamics. *SIAM Journal on Mathematical Analysis*, 33(4):847–859.